

Chapter one

1.0 Manufacturing Processes and Classification

Manufacturing processes are the steps through which raw materials are transformed into a product. The manufacturing processes can be broadly classified into three categories viz. shaping, joining and finishing processes as shown schematically in Figure 1.1. The selection of a particular process from a wide range of choices for a given application requires a hierarchical classification of the processes. For example, Figure 1.2 depicts how the shaping family can be expanded in different classes such as casting, deformation, moulding, composite and powder processing, and prototyping. Next, moulding as a class can be enlarged into a number of member processes such as compression, rotational, transfer, injection moulding, etc. Lastly, each member process can be identified with a number of attributes, which would facilitate the selection of a member process for a given material, dimension, level of requisite tolerances and so on. Similarly, Figure 1.3 depicts how the joining and machining family can be expanded in different classes and actual processes. A brief description of the three broad categories of the manufacturing processes and the corresponding classifications are outlined in the following.

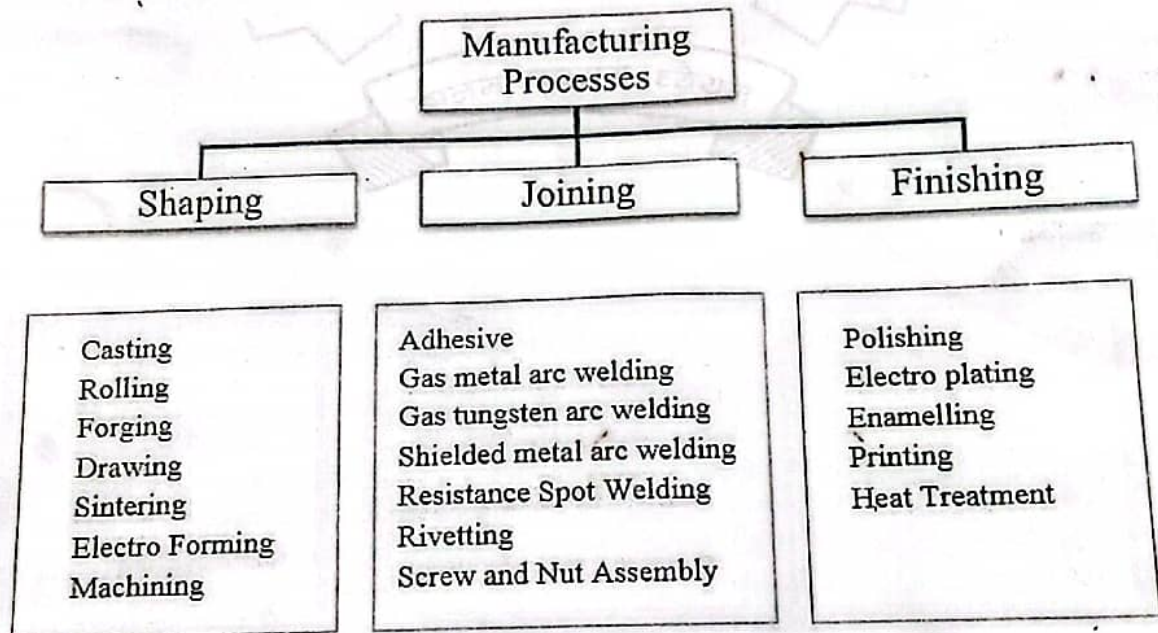


Figure 1.1 Different classes of manufacturing processes

1.2 Shaping Processes

The shaping processes are referred to those that use a certain raw material and shape it to a final part. Casting, moulding, powder material processing, primary and secondary material forming, machining are typical examples of shaping processes. A short exposure of different shaping processes is enlisted below.

1.2.1 Casting Processes

Most of the manufactured parts start its journey with casting process. In a typical casting process, metal is first heated in a furnace until it melts and then the molten metal is poured into a mold so that the liquid metal takes the shape of the mold cavity, which is the final shape of the part. Once the liquid metal in the mold cavity solidifies, the mold is broken or opened to take the final part out of the mold cavity. The metal casting process involves three sequential steps –

- liquefying of metallic material by properly heating it in a suitable furnace,
- pouring of hot molten metal into a previously made colder mould cavity,
- extraction of the solidified cast from the mould cavity

Though casting is one of the oldest manufacturing processes, it is still preferred over other processes due to several advantages listed below:

- It is economical with very little wastage. Even the extra metal produced during each casting can be re-melted and reused.
- It can produce parts with complex geometrical features such as internal cavities, hollow sections with fair dimensional accuracy.
- Casting can be used to make very small to extremely large and complex parts.
- It is possible to cast metallic materials with very low to reasonably high melting temperatures. Further, the mechanical properties of a cast are usually isotropic.

1.3 Classification of casting processes

The casting processes can be classified into two broad categories – expendable mold casting processes and permanent mold casting processes. In expendable mold casting processes, the mold is usually broken to free the solidified cast whereas the mold can be reused in case of permanent mold casting. The pattern used to prepare the molds can also be permanent and expendable, and subsequently, the expendable mold casting processes are further categorized as expendable-pattern-expendable-mold and permanent-pattern-expendable-mold processes. Figure below depicts a detailed classification of the casting processes.

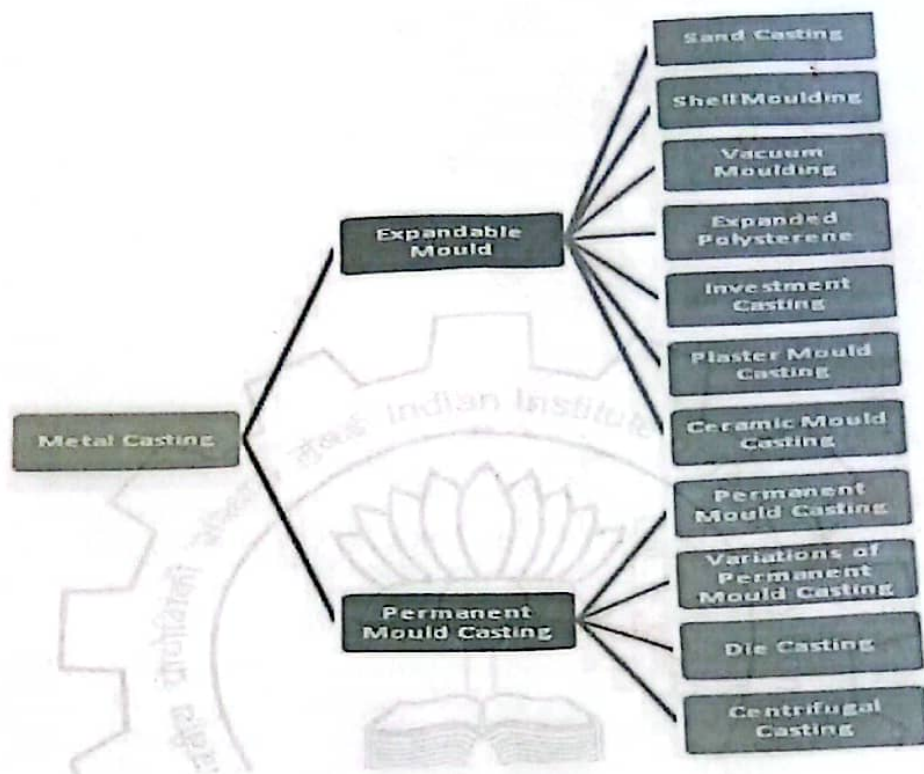


Figure 1.2 Classification of casting processes

Figure below shows a typical mold arrangement for sand casting. Further details of different casting process are discussed in the subsequent lectures.

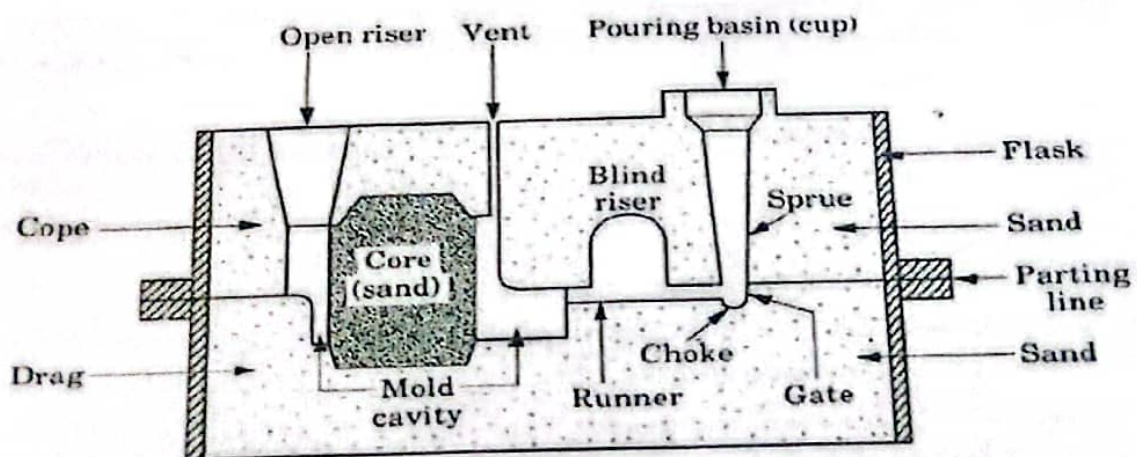


Figure 1.3 Typical mold setup for sand casting process

1.4 Bulk Deformation Processes

The deformation processes shape a final part by plastic deformation with the application of pressure and with or without the assistance of heat. The deformation processes are also referred to as metal forming processes. The metal forming processes that induce a significant shape change starting with a bulk shape rather than sheet are categorized under the bulk deformation processes. In most of the cases cylindrical bars and billets, rectangular billets and slabs, and similar shapes are processed by stressing metal sufficiently in cold, warm or hot condition to cause plastic flow into desired shape. Complex shapes with good mechanical properties can be produced easily and expensively using these processes. Forging, rolling, extrusion and drawing are the common example of bulk deformation process. Depending on the temperature at which the deformation is carried out, these processes can be classified in to hot working and cold working processes. When the plastic deformation is carried out above the recrystallization temperature of the material, the corresponding operation is referred to as hot working. Similarly, when the plastic deformation is induced below the recrystallization temperature of the material, the processes are referred to as cold working. Figure below shows depicts some of the bulk deformation processes in a schematic manner.

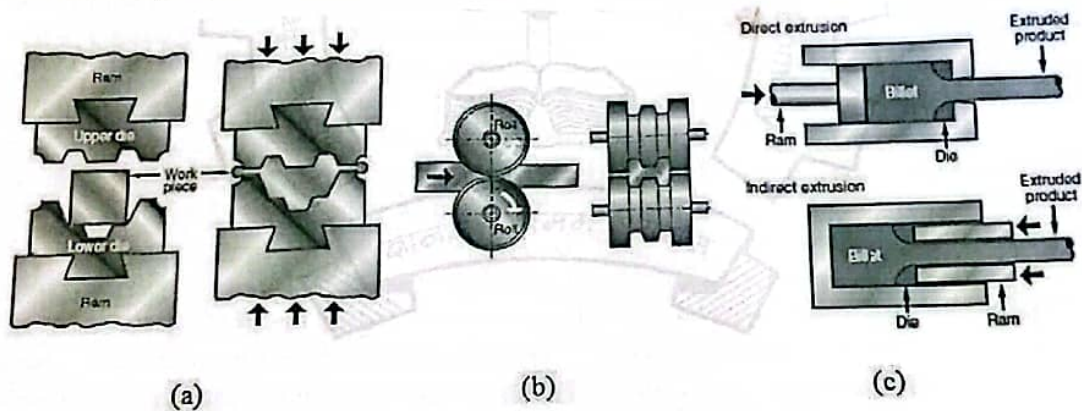


Figure 1.4 Schematic picture of (a) forging, (b) rolling, and (c) rolling processes

1.5 Machining

Machining is a form of subtractive manufacturing in which a sharp cutting tool is used to physically remove material to achieve a desired geometry. Most of the engineering components such as gears, bolts, screws, nuts need dimensional and form accuracy for serving their purpose, which cannot be obtained through casting or deformation process like forging, rolling, etc. Figure below schematically illustrates the basic principle of machining.

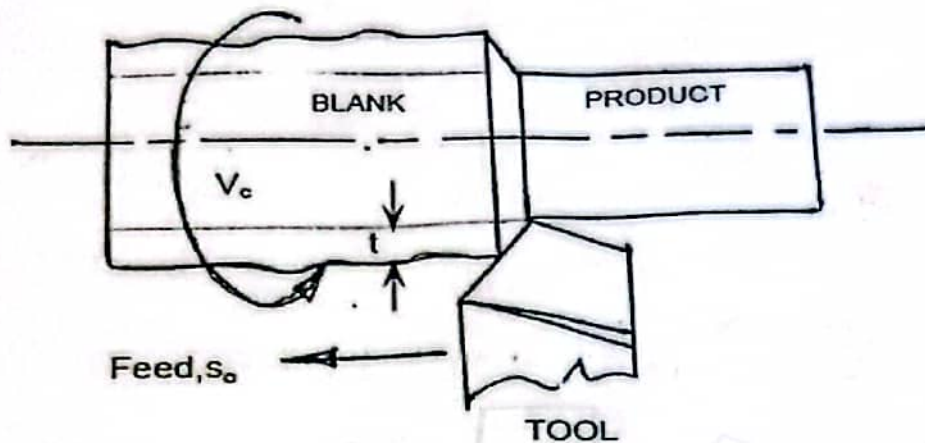


Figure 1.5 Schematic illustration of machining process

A wide variety of machining processes are available today that can broadly be classified in three main categories – conventional machining processes that are used for all kinds of bulk material removal operations, grinding processes that are primarily employed to obtain a desired surface finish, non-conventional or advanced machining processes that are used for special kind of material removal operations. As per the name suggests, non-conventional machining processes do not follow the principle of relative hardness as conventional machining, where the tool material must be harder than the work material for proper removal of material. The processes that remove material by melting, evaporation, chemical and / or electrochemical action etc. are generally referred to as non-conventional machining processes. Electro discharge machining, electrochemical machining, laser and electron beam machining are some of the common examples of non-conventional machining processes. Figure below depicts schematic presentation of various machining operations. The advantages of machining process are manifold. Some of these broad merits of machining processes are listed below.

1. The machining processes can produce a wide variety of dimensions with fine form accuracy.
- 2) Almost all kind of engineering materials and plastics can be machined,
- 3) The machining processes can be easily automated to achieve an excellent productivity,
- 4) The role of the process parameters and their control to obtain a desired part with good dimensional accuracy are well established in most of the machining processes.

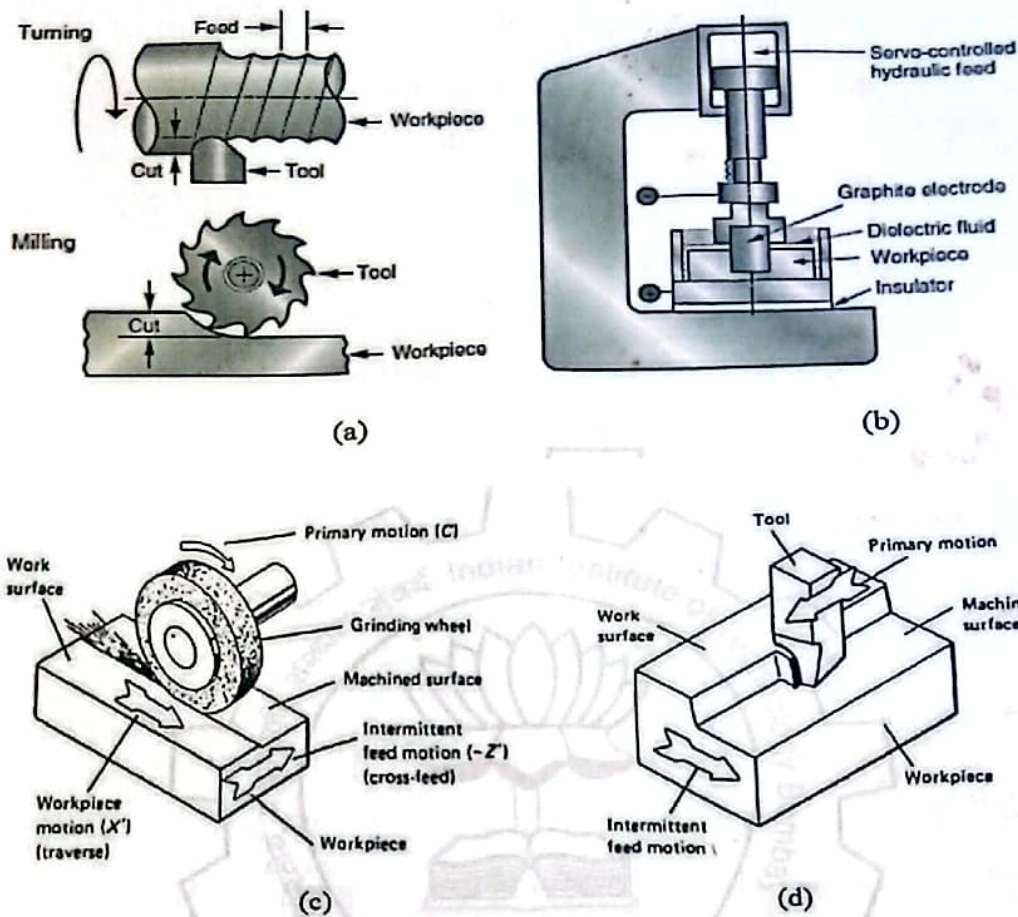


Figure 1.6 Schematic presentations of four machining processes – (a) Turning and milling, b) electro discharge machining, c) surface grinding, d) shaping

1.6 Joining Processes and Surface Treatment Methods

Though shaping is one of the most important processes and can produce a wide range of components, it is often deficient in making a complete product due to the complexity associated with the shape and functions. So, joining of simple pre-shaped parts into a fully functional structure is necessary. Welding, brazing and soldering, fastening, and adhesive bonding are the most commonly used process in joining. Even after joining, further processing may be required to enhance the mechanical properties and aesthetics of the final assembled part or component to meet specific operational conditions. The joining processes can be broadly classified into two groups – one that produces non-permanent joints and the other that constructs permanent joints.

The first group includes some of the common mechanical joining processes such as screws and bolts, snap fits, shrink fits etc. The permanent joining processes can be classified into four groups including mechanical, solid state, liquid state, and liquid-solid state. The mechanical joining processes, which are permanent in nature, include rivet, stitch, staple, and lap-seam. In principle, these joints are heterogeneous in nature since no atomic bonding takes place along the original joint interface.

1.7 Mechanical Joining Processes

The permanent joining processes, which are mechanical in nature, are, in principle, derivatives of the basic metal working processes. These are often referred to fasteners. The most common mechanical joining methods are rivet, nut and bolts, staple, seam joint etc. Figure below schematically depicts a number of mechanical joints.

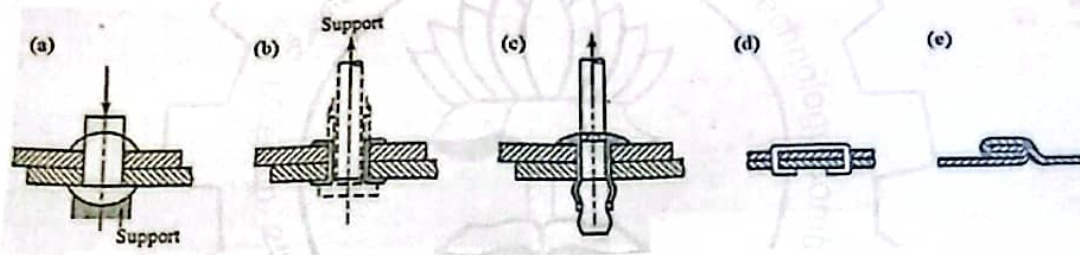


Figure 1.7 Schematic presentation of five mechanical joints: (a) rivet, (b) tubular rivet (c) blind rivet, (d) staple, (e) seam

1.8 Solid State Joining Process

In solid state joining processes, the bonding between the assembled members occurs through adhesion and / or diffusion across the original joint interface. However, adhesions between the two surfaces are difficult due to the presence of surface contaminants such as oxide layers, adsorbed gas films, residual lubricants, etc. Various techniques are adopted to promote the adhesion between two surfaces such as:

- Relative movement of faying surfaces under an axial force that helps to break up surface films facilitating the exposure and mating of clean surfaces,
- Plastic deformation of the contacting bodies leading to growth and extension of the contacting surfaces that would result in rupture of interfacial contaminants and exposure of fresh, clean surfaces, subsequently, creating a solid-state weld,
- Softening of contacting interfaces by localized heating, applied externally or generated in-process, to promote easy plastic deformation and / or inter-atomic diffusion creating a solid-state bond.

Various solid-state joining processes are developed following some of the principles mentioned above. Figure below schematically depicts three common solid state joining processes.

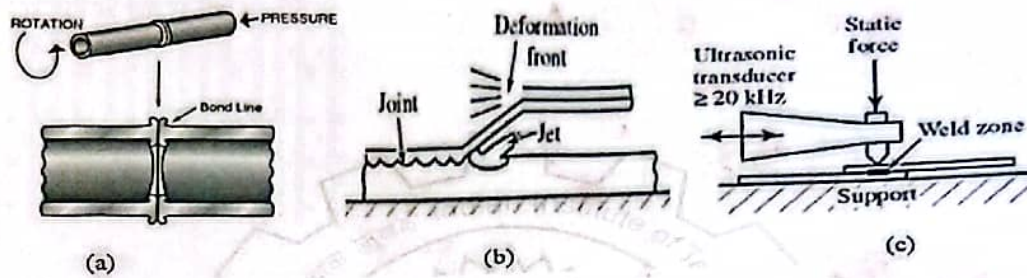


Figure 1.8 Three different types of solid state joining process; (a) Friction welding, (b) Explosion welding, and (c) Ultrasonic welding

1.9 Liquid State Joining Process

The liquid state joining processes involve localized melting and solidification of workpiece materials with or without the addition of external filler material. The liquid state joining processes are commonly referred to as fusion welding. Based on the characteristics of the external filler (electrode) material, the welding processes can also be classified as consumable electrode and non-consumable electrode welding processes. Some of the common fusion welding processes are listed below.

1.9.1 Shielded metal arc welding process (SMAW)

Shielded Metal Arc Welding (SMAW) Process is a manual welding process where an electric arc is used for localized melting of workpiece materials. The electric arc is created between a consumable electrode and the workpiece material as shown schematically in Figure shown below. The electrode is covered with a coating (referred to as flux), which is extruded on the surface of the electrode. During welding, the electrode coating decomposes and melts providing a protective atmosphere on the weld area to avoid the reaction between hot liquid metal and atmospheric gases. SMAW process is the most popular amongst all other fusion arc welding processes since the equipment is inexpensive and easy to use. SMAW process can be used on carbon steels, low alloy steels, stainless steels, cast irons, copper, nickel, and aluminium.

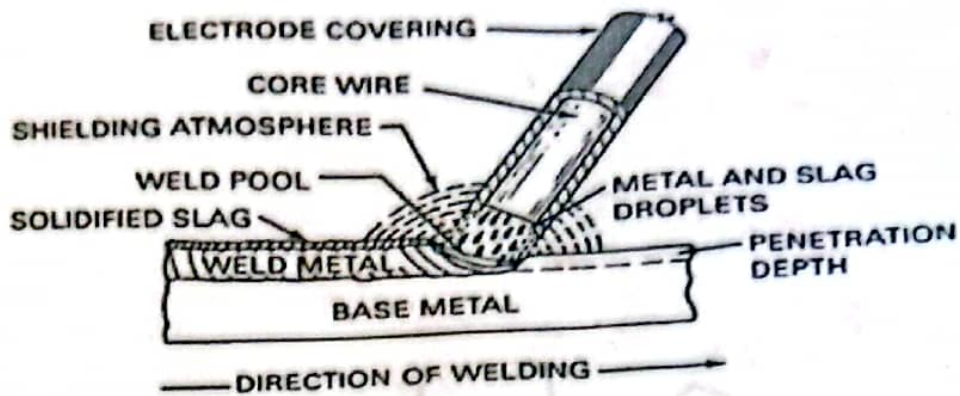


Figure 1.9 Schematic set-up for shielded metal arc welding process

1.9.2 Gas metal arc welding process (GMAW)

In this process, an electric arc is struck between a continuously fed consumable electrode wire and the workpiece material as shown in Figure below. The electrode wire is automatically fed from a spool into the weld pool by a wire feed system. The wire feeder draws the electrode through the welding torch. The shielding is supplied by an inert gas which flows around the wire through a gas cap attached to the torch.

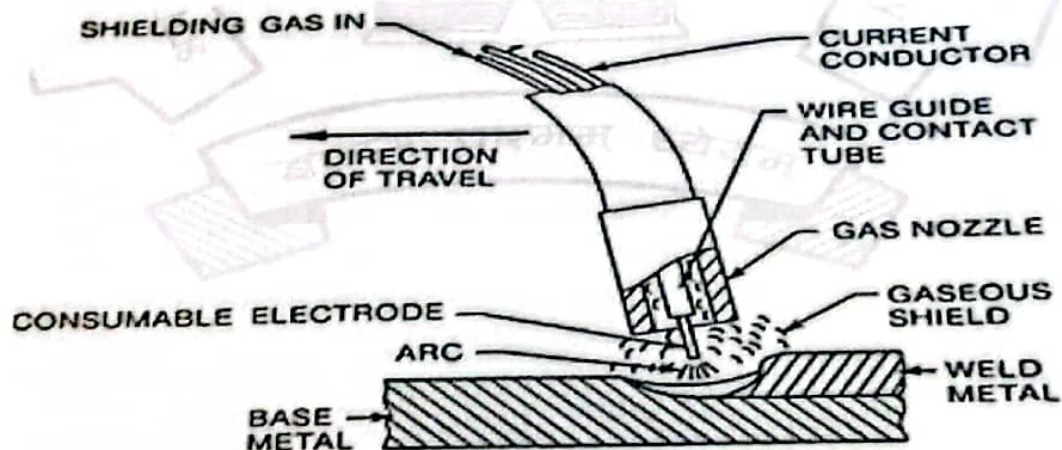


Figure 1.10 Schematic set-up for gas metal arc welding process

1.9.3 Submerged arc welding process (SAW)

Submerged arc welding is almost similar to gas metal arc welding process to the fact that an electric arc is created between a continuously fed consumable solid or tubular electrode wire and workpiece materials. However, the arc and the molten weld pool is submerged under a blanket of granular fusible flux consisting of lime, silica, manganese oxide, calcium fluoride, and other compounds, which are being fed continuously ahead of the welding electrode. Furthermore, the molten flux being electrically conductive provides a current path between the electrode and the workpiece. The thick layer of flux completely covers the molten metal thus preventing spatter and sparks. The process is simple to mechanize and easily automated, and can be used on a wide variety of materials.

1.9.4 Gas tungsten arc welding process (GTAW)

In the gas tungsten arc welding (GTAW) process, the arc is established between the tip of a non-consumable tungsten electrode and the workpiece. Often an extra filler material is used if joint filling is important. Because the arc is created between a shaped (conical) non-consumable tungsten electrode and the workpiece, GTAW process provides a concentrated arc leading to very high heat energy density. An inert shielding gas protects the molten weld pool and the non-consumable tungsten electrode.

1.9.5 Plasma arc welding process (PAW)

The objective of the Plasma Arc Welding (PAW) process is to increase the intensity of the arc plasma in a controlled manner such that greater thickness can be welded with the minimum spread of the welding arc. This objective is achieved by providing a special gas nozzle around a tungsten electrode operating on direct current electrode negative (DCEN) polarity. The constricted plasma formed is highly ionized and concentrated. Two variants of the Plasma Arc Welding (PAW) process are commonly used. One is referred to as the transferred arc process and the second is referred to as the non-transferred arc process. In the transferred arc process, the workpiece is connected to the negative polarity so that the arc plasma irradiates on the workpiece has greater intensity. In the non-transferred arc process, the gas nozzle forms part of the electric circuit and hence, the arc plasma is formed between the tungsten electrode and the nozzle. The tail of the arc plasma that irradiates on the workpiece is therefore of lesser intensity.

1.10 Surface Treatment Methods

The surface treatment processes are used to improve the properties of the surface only. In many applications, it is necessary to harden the surface to prevent abrasive wear. Different types of hardening methods such as quenching, induction hardening, carburizing, nitriding, physical vapour deposition (PVD), chemical vapour deposition (CVD) are some of the commonly used surface treatment method. Similarly, thermal coating like cladding, thermal spraying, hot dipping

etc. surface treatment processes such as painting, electrolytic coating, etc. and conversion coating like oxidising, phosphating, chromating etc. are also employed to improve surface properties.

1.11 Strategy for Selecting Proper Processes

Out of numerous manufacturing processes only a few are listed above. The selection of an appropriate manufacturing process for a given material, requisite shape, dimensional accuracy and service is critical and broadly requires the following steps to follow.

1. Consider all processes as probable candidates
2. Screen them
 - compatibility to material and / or shape and / or precision
3. Rank them using objective
 - economic batch size-ranges
 - relative cost
4. Seek for supporting information for the top ca